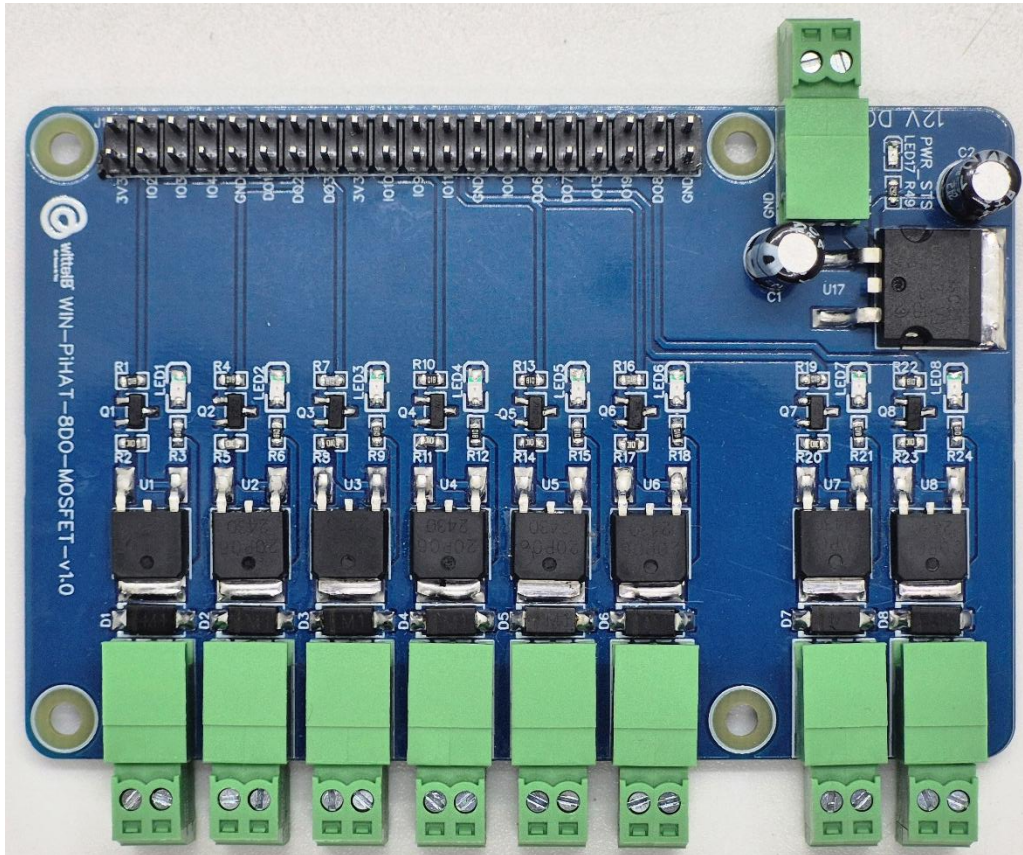


Technical Datasheet of WIN-PIHAT-8DO(MOSFET)

1. Overview

The **Digital Output (DO) (MOSFET) PIHAT** is an 8-channel expansion board for Raspberry Pi 4 and CM4, offering **MOSFET-based digital output** for **industrial load control**. The MOSFET-based output stage protects your system by providing **level shifting and load buffering** between external loads and the Pi's GPIO header, ensuring reliable operation in industrial environments.



PIHAT features a **scalable architecture** supporting **1 to 8 independent MOSFET digital outputs (DO)**. Its **population-based design** allows channel flexibility while maintaining a **compact and reliable solution** for embedded control, automation, and industrial load switching applications.

Specifications

Category	8DO(MOSFET) PIHAT
Output Type	Solid-State Digital Outputs (High-Side), P-Channel MOSFET, 2A @ 12VDC
Channel Count	8 Channels (Expandable Design Architecture)
Connectors	9 of 2.54 mm Pluggable Terminal Blocks (2-Pin), 40-Pin Raspberry Pi GPIO Header

General :-

Dimensions	56 mm L x 85 mm W
Power	Input Power – 12/24V DC
Operating Temperature	0 – 60° C (32 ~ 140°F)
Storage Temperature	-20 - 70° C (-4 ~ 158°F)
Storage Humidity	5 ~ 95 % RH, non – Condensing

Certifications



DO (Digital Outputs) with Relay :-

Channels	8
Output Type	High-Side Switch (P-MOSFET)
Max Load Current	2A Continuous (per channel)
Switching Voltage	12V - 24V DC
Switching Time (On/Off)	< 100 μ s
On Resistance	< 100 Ω
Operation Life	Virtually Unlimited (Solid-State)
Protection	Integrated Flyback Diode

Configuration :-

All 8 DO channels operate as MOSFET-driven digital outputs and are interfaced to the Raspberry Pi GPIO header through a Bipolar Transistor (BJT) driver stage. Output control is performed via software running on the Raspberry Pi. Each channel is configured as a GPIO output and drives a corresponding High-Side MOSFET switch.

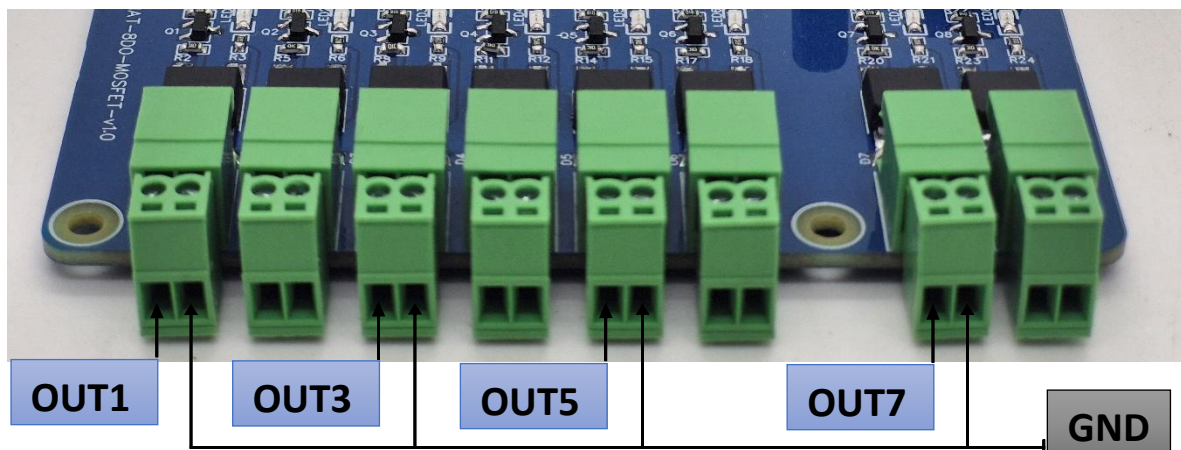
When a GPIO pin is driven HIGH, the Bipolar Transistor pulls the MOSFET Gate LOW, turning the channel ON and supplying power to the load. When the GPIO pin is driven LOW, the MOSFET turns OFF and the output returns to its default state.

The following are step by step guides for configuration:

1. Connect the Module to +12/24V DC POWER SUPPLY.
2. Connect the Board to the Raspberry Pi via header pins

Sr. No.	Digital Input	Pi GPIO	Header Pin number
1	D01	17	11
2	D02	27	13
3	D03	22	15
4	D04	24	18
5	D05	25	22
6	D06	05	29
7	D07	06	31
8	D08	26	37

Wiring Diagram:-



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- When a GPIO pin is driven HIGH, the Bipolar Transistor pulls the MOSFET Gate LOW, turning the channel ON and supplying power to the load. When the GPIO pin is driven LOW, the MOSFET turns OFF and the output returns to its default state.

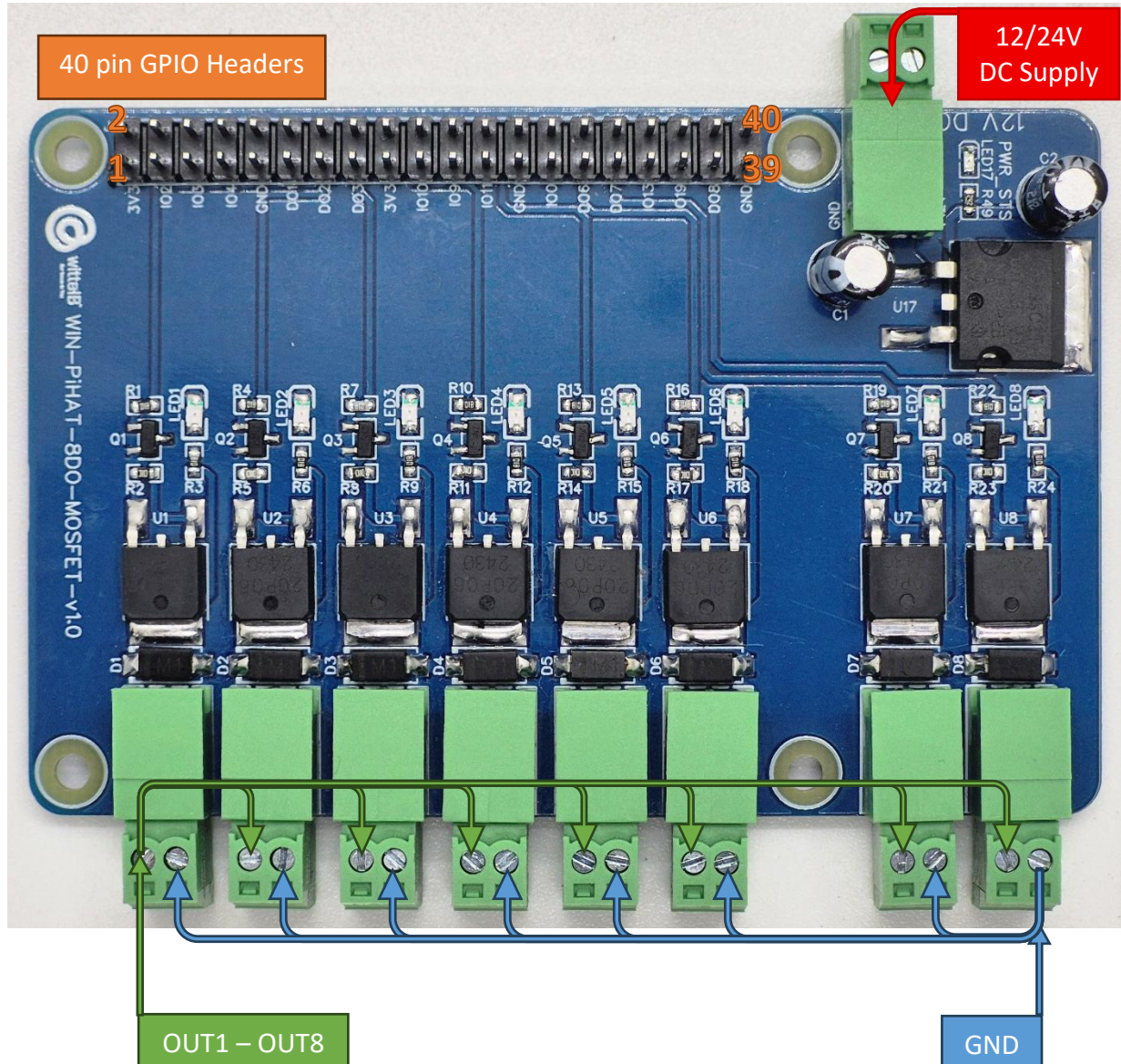
Digital Output Characteristics:

- The DO pin is a high-side output and reflects the input supply voltage.
- Example: Supplying 24V will result in a 24V output, which can damage 12V-rated relays. Ensure compatibility with the load.

Sample Test Code:

- [WIN-PIHAT-8DO\(MOSFET\)](#)

Hardware Interface & Connector Layout:



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